VF Asia Footwear Laboratory Testing Manual

Test Method for Water Absorption – Static	**Test method # VF- E10** **REF # : TM-R-034** **Page 1 of 2**
Date Issued : July 12, 2014 **Revision # :**	**Created by : Nila Babia Erazo/ Chris Adam**

PURPOSE AND SCOPE: To measure the amount of water absorbed by a test material as a result of being immersed in water for a specified period of time.

REFERENCE: ASTM D 1815, SATRA TM 6

EQUIPMENT AND MATERIAL:

1. Immersion dish which is large enough to contain the test sample such that no portion of the test sample contacts the sides of the dish and the test sample is completely immersed in the water.
2. A balance capable of weighing to 100 grams +/-0.001 grams.
3. Blotting paper
4. Piece of glass rod for supporting and weighting the test sample.
5. Steel scale capable of measuring to +/-0.1cm.
6. Leather, fabric, and foam

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Die or hand cut a test sample which is 5.0+/-0.5cm by 5.0+/-0.5cm. For foam material (e.g. midsoles) which will be used in the shoe with the “skin on” a complete midsole unit or test plate with the “skin on” oven the entire surface area is used.

REPETITIONS: Test two (2) samples per material**.**

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Weigh the test sample to the nearest 0.001g (W1). Place the test sample in the immersion dish, support the test samples by placing a glass rod underneath each end of the test sample, weigh the test material by placing a glass rods on the surface of the test material and perpendicular to the supporting glass rods.

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2. Carefully pour distilled water at 23+/-2ºC into the immersion dish until the test sample is completely immersed ,use at least 10 grams of water for every gram which the test sample weighs
3. Allow the test sample to remain immersed in water for :
   Option 1: 60+/-5 minutes
   Option 2: 120+/-5 minutes
4. Remove the test sample from the water ,and lightly blot the surface of the sample to remove any surface water
5. Immediately weigh the test material to the nearest 0.001 grams (W2).

CALCULATION:

$$\text{Water absorption (\%):} = \frac{W2-W1}{W1} * 100$$

REPORT: The average of the two measurements to the nearest 0.1%.